

Command	Description
ls	List files and directories
pwd	Print current working directory
date	Display current date and time
whoami	Display current username
hostname	Display system hostname
uname	Display operating system name
uname -r	Display kernel version (<i>requires argument</i>)
id	Display user and group IDs
tty	Display terminal name
uptime	Show system uptime
users	Show logged-in users
who	List logged-in users
w	Show logged-in users and activity
groups	Display user's groups
env	Display environment variables
printenv	Print environment variables
df	Display disk space usage
free	Show memory usage
ps	Display running processes
top	Display running processes interactively (press q to quit)
vmstat	Display virtual memory statistics
iostat	Display CPU and I/O statistics (if installed)
lscpu	Show CPU information
lsblk	List block storage devices
mount	Display mounted file systems
history	Command history (<i>shell built-in; won't work with exec()</i>)
clear	Clear the terminal screen
echo	Print text (<i>needs an argument</i>)
bc	Calculator (if installed)

Command	Description
factor	Display prime factors of a number (<i>needs an argument</i>)
yes	Print "y" continuously (stop with Ctrl + C)
sleep	Pause execution (<i>needs a time argument</i>)

```

varshini@DESKTOP-BLT5L3R:~$ ls
OSSP  fork_exec.c  hello  hello.c  process.c
varshini@DESKTOP-BLT5L3R:~$ pwd
/home/varshini
varshini@DESKTOP-BLT5L3R:~$ date
Mon Aug  3 05:31:47 UTC 2026
varshini@DESKTOP-BLT5L3R:~$ whoami
varshini
varshini@DESKTOP-BLT5L3R:~$ hostname
DESKTOP-BLT5L3R
varshini@DESKTOP-BLT5L3R:~$ uname
Linux
varshini@DESKTOP-BLT5L3R:~$ uname -r
6.18.33.2-microsoft-standard-WSL2
varshini@DESKTOP-BLT5L3R:~$ id
uid=1000(varshini) gid=1000(varshini) groups=1000(varshini),27(sudo),100(users)
varshini@DESKTOP-BLT5L3R:~$ tty
/dev/pts/0
varshini@DESKTOP-BLT5L3R:~$ uptime
 05:32:48 up 48 min,  1 user,  load average: 0.04, 0.04, 0.00
varshini@DESKTOP-BLT5L3R:~$ users
varshini
varshini@DESKTOP-BLT5L3R:~$ who
varshini pts/1        2026-08-03 04:42
varshini@DESKTOP-BLT5L3R:~$ w
 05:33:01 up 49 min,  1 user,  load average: 0.03, 0.03, 0.00
USER      TTY      FROM          LOGIN@      IDLE        JCPU   PCPU   WHAT
varshini  pts/1    -             04:42      50:15      0.04s   0.02s  -bash
varshini@DESKTOP-BLT5L3R:~$ groups
varshini sudo users

```

```
varshini@DESKTOP-BLT5L3R:~$ env
SHELL=/bin/bash
WSL2_GUI_APPS_ENABLED=1
WSL_DISTRO_NAME=Ubuntu
NAME=DESKTOP-BLT5L3R
PWD=/home/varshini
LOGNAME=varshini
HOME=/home/varshini
LANG=C.UTF-8
WSL_INTEROP=/run/WSL/981_interop
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:
ni=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:
*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo
=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz
=01;31:*.zst=01;31:*.tztst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31
:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo
=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:
*.avif=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:
*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg
=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;3
5:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:
*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01
;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.og
x=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;
36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*
=00;90:.*#=00;90:*.bak=00;90:*.crdownload=00;90:*.dpkg-dist=00;90:*.dpkg-new=00;90:*.dpkg-old=00;90:*.d
pkg-tmp=00;90:*.old=00;90:*.orig=00;90:*.part=00;90:*.rej=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.rpmsa
ve=00;90:*.swp=00;90:*.tmp=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:
WAYLAND_DISPLAY=wayland-0
LESSCLOSE=/usr/bin/lesspipe %s %s
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=varshini
DISPLAY=:0
SHLVL=1
XDG_RUNTIME_DIR=/run/user/1000
WLENV=
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib
/wsl/lib:/mnt/c/Python34/Scripts:/mnt/c/Python34:/mnt/c/Program Files/Eclipse Adoptium/jdk-21.0.9
.10-hotspot/bin:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Comm
n Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Wind
ows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1
0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/Java/jdk-21/bin:
/mnt/c/ProgramData/Chocolatey/bin:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/MySQL/MySQL Shell
8.0/bin:/mnt/c/Users/Varshini/.local/bin:/mnt/c/Users/Varshini/AppData/Local/Microsoft/WindowsApps:/m
nt/c/Users/Varshini/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/Varshini/AppData/Roaming
/npm:/mnt/c/Users/Varshini/PyCharm 2026.1.2/bin:/snap/bin
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
HOSTTYPE=x86_64
PULSE_SERVER=unix:/mnt/wslg/PulseServer
_=usr/bin/env
```

```
varshini@DESKTOP-BLT5L3R:~$ printenv
SHELL=/bin/bash
WSL2_GUI_APPS_ENABLED=1
WSL_DISTRO_NAME=Ubuntu
NAME=DESKTOP-BLT5L3R
PWD=/home/varshini
LOGNAME=varshini
HOME=/home/varshini
LANG=C.UTF-8
WSL_INTEROP=/run/WSL/981_interop
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:
mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:
*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo
=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz
=01;31:*.zst=01;31:*.tztst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31
:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo
=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:
*.avif=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:
*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg
=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;3
5:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:
*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01
;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.og
x=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;
36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*~
=00;90:*#=00;90:*.bak=00;90:*.crdownload=00;90:*.dpkg-dist=00;90:*.dpkg-new=00;90:*.dpkg-old=00;90:*.d
pkg-tmp=00;90:*.old=00;90:*.orig=00;90:*.part=00;90:*.rej=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.rpmsa
ve=00;90:*.swp=00;90:*.tmp=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:
WAYLAND_DISPLAY=wayland-0
LESSCLOSE=/usr/bin/lesspipe %s %s
TERM=xterm-256color
LESSOPEN=| /usr/bin/lesspipe %s
USER=varshini
DISPLAY=:0
SHLVL=1
XDG_RUNTIME_DIR=/run/user/1000
WLENV=
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/snapd/desktop
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib
/wsl/lib:/mnt/c/Python314/Scripts:/mnt/c/Python314:/mnt/c/Program Files/Eclipse Adoptium/jdk-21.0.9.
10-hotspot/bin:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Commo
n Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Wind
ows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsPowerShell/v1.
0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/Java/jdk-21/bin:
/mnt/c/ProgramData/chocolatey/bin:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/MySQL/MySQL Shell
8.0/bin:/mnt/c/Users/Varshini/.local/bin:/mnt/c/Users/Varshini/AppData/Local/Microsoft/WindowsApps:/m
nt/c/Users/Varshini/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/Varshini/AppData/Roaming
/npm:/mnt/c/Users/Varshini/PyCharm 2026.1.2/bin:/snap/bin
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus
HOSTTYPE=x86_64
PULSE_SERVER=unix:/mnt/wslg/PulseServer
_=usr/bin/printenv
```

```
varshini@DESKTOP-BLT5L3R:~$ df
Filesystem      1K-blocks      Used Available  Use% Mounted on
none            3985736          0   3985736    0% /usr/lib/modules/6.18.33.2-micro
soft-standard-WSL2
none            3985736          4   3985732    1% /mnt/wsl
drivers         269479932 159097884 110382048   60% /usr/lib/wsl/drivers
/dev/sdd         1055762868 2295132 999764264    1% /
none            3985736          36   3985700    1% /mnt/wslg
none            3985736          0   3985736    0% /usr/lib/wsl/lib
rootfs          3979976      2772   3977204    1% /init
none            3985736          524   3985212    1% /run
none            3985736          0   3985736    0% /run/lock
none            3985736          0   3985736    0% /run/shm
none            3985736          80   3985656    1% /mnt/wslg/versions.txt
none            3985736          80   3985656    1% /mnt/wslg/doc
C:\              269479932 159097884 110382048   60% /mnt/c
D:\              104856572  167640 104688932    1% /mnt/d
E:\              104856572  841172 104015400    1% /mnt/e
tmpfs            797144         20    797124    1% /run/user/1000
varshini@DESKTOP-BLT5L3R:~$ free
              total        used        free      shared  buff/cache   available
Mem:          7971472      543404      7354440         3576      228024      7428068
Swap:         2097152           0       2097152
varshini@DESKTOP-BLT5L3R:~$ ps
  PID TTY          TIME CMD
   987 pts/0    00:00:00 bash
  1077 pts/0    00:00:00 ps
varshini@DESKTOP-BLT5L3R:~$ |
```

```
varshini@DESKTOP-BLT5L3R:~$ top
top - 05:36:45 up 52 min, 1 user, load average: 0.01, 0.02, 0.00
Tasks: 23 total, 1 running, 22 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 99.9 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  7784.6 total,  7180.9 free,   531.5 used,   223.0 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used.  7253.2 avail Mem

   PID USER      PR  NI  VIRT  RES  SHR S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   21904 13116  9720 S   0.0   0.2   0:00.97 systemd
    2 root        20   0   3180  2212  2072 S   0.0   0.0   0:00.02 init-syste+
    8 root        20   0   3196  2128  2008 S   0.0   0.0   0:00.00 init
   54 root        19  -1  50368 16328 15228 S   0.0   0.2   0:00.45 systemd-jo+
  104 root        20   0   25416 6684  5128 S   0.0   0.1   0:00.74 systemd-ud+
  156 systemd+    20   0  21344 13180 11056 S   0.0   0.2   0:00.13 systemd-re+
  159 systemd+    20   0  91036  7988  7024 S   0.0   0.1   0:00.19 systemd-ti+
  174 root        20   0   4244  2680  2432 S   0.0   0.0   0:00.01 cron
  175 message+    20   0   9608  5416  4692 S   0.0   0.1   0:00.24 dbus-daemon
  191 root        20   0  17980  8836  7792 S   0.0   0.1   0:00.21 systemd-lo+
  195 root        20   0 1755848 11512  9000 S   0.0   0.1   0:00.27 wsl-pro-se+
  198 syslog      20   0  222516  6024  4616 S   0.0   0.1   0:00.13 rsyslogd
  215 root        20   0   3124  1976  1836 S   0.0   0.0   0:00.02 agetty
  233 root        20   0 107032 23252 13796 S   0.0   0.3   0:00.14 unattended+
  312 root        20   0   6704  4664  3876 S   0.0   0.1   0:00.02 login
  358 varshini    20   0  20156 11400  9452 S   0.0   0.1   0:00.12 systemd
  359 varshini    20   0  21168  3576  1848 S   0.0   0.0   0:00.00 (sd-pam)
  381 varshini    20   0   6080  5324  3660 S   0.0   0.1   0:00.02 bash
  717 polkitd      20   0 308172  8092  7196 S   0.0   0.1   0:00.15 polkitd
  979 root        20   0   3184  1124   980 S   0.0   0.0   0:00.00 SessionLea+
  981 root        20   0   3200  1260  1108 S   0.0   0.0   0:00.03 Relay(987)
  987 varshini    20   0   6080  5396  3676 S   0.0   0.1   0:00.11 bash
 1080 varshini    20   0   9308  5708  3512 R   0.0   0.1   0:00.01 top
```

```
varshini@DESKTOP-BLT5L3R:~$ vmstat
procs -----memory----- --swap-- -----io---- -system-- -----cpu-----
 r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs us sy id wa st gu
  1  0     0 7354004  7808 220596      0    0   311    59  162    0  0  0 100  0  0  0
varshini@DESKTOP-BLT5L3R:~$ iostat
Command 'iostat' not found, but can be installed with:
sudo apt install sysstat
```

```

varshini@DESKTOP-BLT5L3R:~$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 12
On-line CPU(s) list:    0-11
Vendor ID:              GenuineIntel
Model name:             13th Gen Intel(R) Core(TM) i5-1334U
CPU family:             6
Model:                  186
Thread(s) per core:     2
Core(s) per socket:     6
Socket(s):              1
Stepping:               3
BogoMIPS:                4992.00
Flags:                  fpu vme de pse tsc msr pae mce cx8 a
                        pic sep mtrr pge mca cmov pat pse36
                        clflush mmx fxsr sse sse2 ss ht sysc
                        all nx pdpe1gb rdtscp lm constant_ts
                        c rep_good nopl xtopology tsc_reliab
                        le nonstop_tsc cpuid tsc_known_freq
                        pni pclmulqdq vmx ssse3 fma cx16 pci
                        d sse4_1 sse4_2 x2apic movbe popcnt
                        tsc_deadline_timer aes xsave avx fl6
                        c rdrand hypervisor lahf_lm abm 3dno
                        wprefetch ssbd ibrs ibpb stibp ibrs_
                        enhanced tpr_shadow ept vpid ept_ad
                        fsgsbase tsc_adjust bmi1 avx2 smep b
                        mi2 erms invpcid rdseed adx smap clf
                        lushopt clwb sha_ni xsaveopt xsavec
                        xgetbv1 xsaves avx_vnni vnm i umip wa
                        itpkg gfni vaes vpclmulqdq rdpid mov
                        dir movdir64b fsrm md_clear seriali
                        ze ibt flush_lld arch_capabilities

Virtualization features:
  Virtualization:        VT-x
  Hypervisor vendor:      Microsoft
  Virtualization type:    full
Caches (sum of all):
  L1d:                    288 KiB (6 instances)
  L1i:                    192 KiB (6 instances)
  L2:                      7.5 MiB (6 instances)
  L3:                      12 MiB (1 instance)
NUMA:
  NUMA node(s):           1
  NUMA node0 CPU(s):      0-11
Vulnerabilities:
  Gather data sampling:    Not affected
  Ghostwrite:              Not affected
  Indirect target selection: Not affected
  Itlb multihit:           Not affected
  L1tf:                    Not affected
  Mds:                     Not affected
  Meltdown:                Not affected
  Mmio stale data:         Not affected
  Old microcode:           Not affected
  Reg file data sampling:   Mitigation; Clear Register File
  Retbleed:                Mitigation; Enhanced IBRS
  Spec rstack overflow:    Not affected
  Spec store bypass:       Mitigation; Speculative Store Bypass
                           disabled via prctl
  Spectre v1:              Mitigation; usercopy/swapgs barriers
                           and __user pointer sanitization
  Spectre v2:              Mitigation; Enhanced / Automatic IBR
                           S; IBPB conditional; PBR SB-eIBRS SW
                           sequence; BHI BHI_DIS_S
  Srbds:                   Not affected
  Tsa:                     Not affected
  Tsx async abort:         Not affected
  Vmscape:                 Not affected
varshini@DESKTOP-BLT5L3R:~$

```

```

varshini@DESKTOP-BLT5L3R:~$ lsblk
NAME
MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda  8:0    0 356.9M 1 disk
sdb  8:16   0 159.4M 1 disk
sdc  8:32   0    2G  0 disk [SWAP]
sdd  8:48   0    1T  0 disk /mnt/wslg/distro
/

varshini@DESKTOP-BLT5L3R:~$ mount
none on /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2 type overlay (rw,nosuid,nodev,noatime,lowerdir=/modules
,upperdir=/lib/modules/6.18.33.2-microsoft-standard-WSL2/rw/upper,workdir=/lib/modules/6.18.33.2-microsoft-standar
d-WSL2/rw/work,uuid=on)
none on /mnt/wsl type tmpfs (rw,relatime)
drivers on /usr/lib/wsl/drivers type 9p (ro,nosuid,nodev,noatime,aname=drivers;fmask=222;dmask=222,cache=0x5,acce
s=client,msize=65536,trans=fd,rfd=8,wfd=8)
/dev/sdd on / type ext4 (rw,relatime,discard,errors=remount-ro,data=ordered)
none on /mnt/wslg type tmpfs (rw,relatime)
/dev/sdd on /mnt/wslg/distro type ext4 (ro,relatime,discard,errors=remount-ro,data=ordered)
none on /usr/lib/wsl/lib type overlay (rw,nosuid,nodev,noatime,lowerdir=/gpu_lib_packaged:/gpu_lib_inbox,upperdir=
/gpu_lib/rw/upper,workdir=/gpu_lib/rw/work,uuid=on)
rootfs on /init type rootfs (ro,size=3979976k,nr_inodes=994994)
none on /dev type devtmpfs (rw,nosuid,relatime,size=3979976k,nr_inodes=994994,mode=755)
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,noatime)
proc on /proc type proc (rw,nosuid,nodev,noexec,noatime)
devpts on /dev/pts type devpts (rw,nosuid,noexec,noatime,gid=5,mode=620,ptmxmode=000)
none on /run type tmpfs (rw,nosuid,nodev,mode=755)
none on /run/lock type tmpfs (rw,nosuid,nodev,noexec,noatime)
none on /run/shm type tmpfs (rw,nosuid,nodev,noatime)
none on /dev/shm type tmpfs (rw,nosuid,nodev,noatime)
none on /run/user type tmpfs (rw,nosuid,nodev,noexec,noatime,mode=755)
binfmt_misc on /proc/sys/fs/binfmt_misc type binfmt_misc (rw,relatime)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,nsdelegate)
none on /mnt/wslg/versions.txt type overlay (rw,relatime,lowerdir=/systemvhd,upperdir=/system/rw/upper,workdir=/sy
stem/rw/work,uuid=on)
none on /mnt/wslg/doc type overlay (rw,relatime,lowerdir=/systemvhd,upperdir=/system/rw/upper,workdir=/system/rw/w
ork,uuid=on)
none on /tmp/.X11-unix type tmpfs (ro,relatime)
C:\ on /mnt/c type 9p (rw,noatime,aname=drvfs;path=C:\;uid=1000;gid=1000;symlinkroot=/mnt/,cache=0x5,access=client
,msize=65536,trans=fd,rfd=6,wfd=6)
D:\ on /mnt/d type 9p (rw,noatime,aname=drvfs;path=D:\;uid=1000;gid=1000;symlinkroot=/mnt/,cache=0x5,access=client
,msize=65536,trans=fd,rfd=6,wfd=6)
E:\ on /mnt/e type 9p (rw,noatime,aname=drvfs;path=E:\;uid=1000;gid=1000;symlinkroot=/mnt/,cache=0x5,access=client
,msize=65536,trans=fd,rfd=6,wfd=6)
none on /run/user type tmpfs (rw,relatime)
hugetlbfs on /dev/hugepages type hugetlbfs (rw,nosuid,nodev,relatime,pagesize=2M)
debugfs on /sys/kernel/debug type debugfs (rw,nosuid,nodev,noexec,relatime)
mqueue on /dev/mqueue type mqueue (rw,nosuid,nodev,noexec,relatime)
tracefs on /sys/kernel/tracing type tracefs (rw,nosuid,nodev,noexec,relatime)
configfs on /sys/kernel/config type configfs (rw,nosuid,nodev,noexec,relatime)
fusectl on /sys/fs/fuse/connections type fusectl (rw,nosuid,nodev,noexec,relatime)
tmpfs on /run/user/1000 type tmpfs (rw,nosuid,nodev,relatime,size=797144k,nr_inodes=199286,mode=700,uid=1000,gid=1
000)
tmpfs on /mnt/wslg/run/user/1000 type tmpfs (rw,nosuid,nodev,relatime,size=797144k,nr_inodes=199286,mode=700,uid=1
000,gid=1000)
varshini@DESKTOP-BLT5L3R:~$

```

```
varshini@DESKTOP-BLT5L3R:~$ history
```

```
 1  sudo apt update
 2  sudo apt install build-essential
 3  gcc --version
 4  ls
 5  pwd
 6  date
 7  whoami
 8  hostname
 9  uname
10  uname-r
11  id
12  tty
13  uptime
14  users
15  who
16  w
17  groups
18  env
19  printenv
20  df
21  free
22  ps
23  top
24  pwd
25  ls
26  pwwd
27  pwd
28  date
29  whoami
30  hostname
31  uname
32  uname-r
33  uname -r
34  id
35  tty
```



```
37 users
38 who
39 w
40 groups
41 env
42 printenv
43 df
44 free
45 ps
46 top
47 vmstat
48 iostat
49 lscpu
50 lsblk
51 lsblk
52 mount
53 history
54 clear
55 echo
56 echo hello
57 bc
58 factor
59 factor 44
60 yes
61 sleep 3:00
62 sleep 5
63 sleep 4
64 hostname
65 whoami
66 sleep 5
67 tty
68 id
69 mkdir OSSP
70 cd OSSP
71 mkdir src include obj bin docs tests scripts assets
72 touch src/main.c src/shell.c src/parser.c src/executor.c src/bu
```

```
75 touch docs/design.md docs/report.pdf
76 touch tests/test_parser.c tests/test_executor.c
77 touch scripts/run.sh
78 ls
79 git init
80 sudo apt install tree
81 tree
82 touch bin/my_shell
83 touch assets/architecture.png
84 git init
85 tree
86 .git/
87 ls -a
88 gcc --version
89 touch bin/my_shell
90 touch assets/architecture.png
91 ls -a
92 tree -a
93 man fork
94 bash
95 ps
96 pstree
97 ps -ef
98 varshini      323      322      -bash
99 nano fork_exec.c
100 gcc fork_exec.c -o fork_exec
101 q
102 man exec
103 gcc fork_exec.c -o fork_exec
104 ./fork_exec
105 cd ~/OSSP
106 nano fork_exec.c
107 gcc fork_exec.c -o fork_exec
108 ./fork_exec
109 Enter a Linux command:
110 Enter a Linux command: ls
111 nano fork_exec.c
112 gcc fork_exec.c -o fork_exec
113 tree
114 ubuntu_V
115 sudo apt install build-essential
116 gcc --version
117 nano process.c
118 nano.process.c
```

```
119 gcc hello.c -o hello
120 nano hello.c
121 gcc hello.c -o hello
122 ./hello
123 clear
124 mkdir OSSP
125 tree
126 cd ~/OSSP
127 nano fork_exec.c
128 gcc fork_exec.c -o fork_exec
129 ./fork_exec
130 nano fork_exec.c
131 sudo apt update
132 sudo apt install build-essential
133 gcc --version
134 nano hello.c
135 gcc hello.c -o hello
136 ./hello
137 ls
138 pwd
139 date
140 whoami
141 hostname
142 uname
143 uname -r
144 id
145 tty
146 uptime
147 users
148 who
149 w
150 groups
151 env
152 printenv
153 df
154 free
155 ps
156 top
157 vmstat
158 iostat
159 lscpu
160 lsblk
161 mount
162 history
```

```
varshini@DESKTOP-BLT5L3R:~$ |
```



```
varshini@DESKTOP-BLT5L3R: ~$
```

```
varshini@DESKTOP-BLT5L3R:~$
```